

MATH 327 Homework 1 Fall 2026

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1. Problem 1.3

Lemma 1. For all $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We proceed by induction on n .

Base case. For $n = 1$,

$$1 = \frac{1(1+1)}{2},$$

so $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$1 + 2 + \cdots + m = \frac{m(m+1)}{2}. \tag{1}$$

Adding $m + 1$ to both sides of (1), we get

$$\begin{aligned} 1 + 2 + \cdots + m + (m + 1) &= \frac{m(m+1)}{2} + (m + 1) \\ &= (m + 1) \left(\frac{m}{2} + 1 \right) \\ &= (m + 1) \left(\frac{m + 2}{2} \right) \\ &= \frac{(m + 1)(m + 2)}{2}. \end{aligned}$$

Therefore, $P(m + 1)$ holds, and the result follows by induction. ■

Theorem 1. For all $n \in \mathbb{N}$,

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

Proof. Let $P(n)$ be the statement

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

We proceed by induction on n .

Base case. For $n = 1$,

$$1^3 = 1^2,$$

so $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$1^3 + 2^3 + \cdots + m^3 = (1 + 2 + \cdots + m)^2. \quad (2)$$

Adding $(m+1)^3$ to both sides of (2), we get

$$\begin{aligned} 1^3 + 2^3 + \cdots + m^3 + (m+1)^3 &= (1 + 2 + \cdots + m)^2 + (m+1)^3 \\ &= \left(\frac{m(m+1)}{2} \right)^2 + (m+1)^3 && \text{by Lemma 1} \\ &= (m+1)^2 \left(\frac{m^2}{4} + m + 1 \right) \\ &= (m+1)^2 \left(\frac{m^2 + 4m + 4}{4} \right) \\ &= (m+1)^2 \left(\frac{(m+2)^2}{4} \right) \\ &= \left(\frac{(m+1)(m+2)}{2} \right)^2 \\ &= (1 + 2 + \cdots + m + (m+1))^2 && \text{by Lemma 1.} \end{aligned}$$

Therefore, $P(m+1)$ holds, and the result follows by induction. ■

2. Problem 1.12(c)

The following supporting identity is part 1.12(b).

Lemma 2 (Pascal identity from 1.12(b)). *For $n \in \mathbb{N}$ and $k = 1, 2, \dots, n$,*

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}.$$

Proof. Using the definition of the binomial coefficients,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{k-1} = \frac{n!}{(k-1)!(n-k+1)!},$$

and

$$\binom{n+1}{k} = \frac{(n+1)!}{k!(n-k+1)!}.$$

Therefore,

$$\begin{aligned}
\binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\
&= n! \left(\frac{n-k+1}{k!(n-k+1)!} + \frac{k}{k!(n-k+1)!} \right) \\
&= n! \left(\frac{n+1}{k!(n-k+1)!} \right) \\
&= \frac{(n+1)!}{k!(n-k+1)!} \\
&= \binom{n+1}{k}.
\end{aligned}$$

■

Theorem 2 (Binomial theorem). *For all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,*

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$

Proof. Let $P(n)$ be the stated binomial identity. We proceed by induction on n .

Base case. Let $n = 1$. Then $P(1)$ states

$$\begin{aligned}
(a+b)^1 &= \binom{1}{0}a + \binom{1}{1}b \\
&= a + b.
\end{aligned}$$

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus, we have

$$\begin{aligned}
(a+b)^m &= \binom{m}{0}a^m + \binom{m}{1}a^{m-1}b \\
&\quad + \binom{m}{2}a^{m-2}b^2 + \dots \\
&\quad + \binom{m}{m-1}ab^{m-1} + \binom{m}{m}b^m \\
&= a^m + ma^{m-1}b + \frac{1}{2}m(m-1)a^{m-2}b^2 \\
&\quad + \dots + mab^{m-1} + b^m.
\end{aligned}$$

In these expanded expressions, only terms with valid indices are present when m is small.

$$\begin{aligned}
(a+b)^{m+1} &= (a+b)^m(a+b) \\
&= (a+b) \left[\binom{m}{0} a^m + \binom{m}{1} a^{m-1} b \right. \\
&\quad + \binom{m}{2} a^{m-2} b^2 + \dots \\
&\quad \left. + \binom{m}{m-1} a b^{m-1} + \binom{m}{m} b^m \right] \\
&= \binom{m}{0} a^{m+1} + \binom{m}{1} a^m b \\
&\quad + \binom{m}{2} a^{m-1} b^2 + \dots \\
&\quad + \binom{m}{m-1} a^2 b^{m-1} + \binom{m}{m} a b^m \\
&\quad + \binom{m}{0} a^m b + \binom{m}{1} a^{m-1} b^2 \\
&\quad + \binom{m}{2} a^{m-2} b^3 + \dots \\
&\quad + \binom{m}{m-1} a b^m + \binom{m}{m} b^{m+1} \\
&= \binom{m}{0} a^{m+1} \\
&\quad + \left(\binom{m}{1} + \binom{m}{0} \right) a^m b \\
&\quad + \left(\binom{m}{2} + \binom{m}{1} \right) a^{m-1} b^2 \\
&\quad + \dots \\
&\quad + \left(\binom{m}{m} + \binom{m}{m-1} \right) a b^m \\
&\quad + \binom{m}{m} b^{m+1}.
\end{aligned}$$

By part (b),

$$\binom{m}{k} + \binom{m}{k-1} = \binom{m+1}{k}.$$

Also,

$$\begin{aligned}
\binom{m}{0} &= \frac{m!}{0!(m-0)!} = 1 = \binom{m+1}{0}, \\
\binom{m}{m} &= \frac{m!}{m!(m-m)!} = 1 = \binom{m+1}{m+1}.
\end{aligned}$$

Hence,

$$\begin{aligned}
(a+b)^{m+1} &= \binom{m+1}{0} a^{m+1} \\
&\quad + \binom{m+1}{1} a^m b \\
&\quad + \binom{m+1}{2} a^{m-1} b^2 + \dots \\
&\quad + \binom{m+1}{m} a b^m \\
&\quad + \binom{m+1}{m+1} b^{m+1} \\
&= a^{m+1} + (m+1) a^m b \\
&\quad + \frac{1}{2} (m+1) m a^{m-1} b^2 \\
&\quad + \dots + (m+1) a b^m + b^{m+1}.
\end{aligned}$$

Thus, $P(m)$ implies $P(m+1)$. By induction, $P(n)$ is true for all $n \in \mathbb{N}$, so the binomial theorem holds. ■

3. Bernoulli's inequality

Theorem 3. For every $x \geq -1$ and every $n \in \mathbb{N}$,

$$(1+x)^n \geq 1+nx.$$

Proof. Fix $x \in [-1, \infty)$, and let $P(n)$ be the statement

$$(1+x)^n \geq 1+nx.$$

We prove $P(n)$ for all $n \in \mathbb{N}$ by induction.

Base case. For $n = 1$,

$$(1+x)^1 = 1+x = 1+1x,$$

so $(P(1))$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$(1+x)^m \geq 1+mx.$$

Since $x \geq -1$, we have $1+x \geq 0$. Multiplying both sides by $(1+x)$ preserves the inequality:

$$\begin{aligned}
(1+x)^{m+1} &\geq (1+mx)(1+x) \\
&= 1+(m+1)x+mx^2.
\end{aligned}$$

Because $m \in \mathbb{N}$ and $x^2 \geq 0$, we have $mx^2 \geq 0$. Hence,

$$(1+x)^{m+1} \geq 1+(m+1)x.$$

Thus $(P(m+1))$ holds, and the result follows by induction. ■

4. Problem 2.4

Lemma 3. $\sqrt{3}$ is irrational.

Proof. Suppose, for the sake of contradiction, that $\sqrt{3}$ is rational. Then

$$\sqrt{3} = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. We may assume that p and q are relatively prime. Squaring both sides gives

$$3q^2 = p^2.$$

We consider the possible parities of p and q .

Case 1: p and q are both odd. Write $p = 2k + 1$ and $q = 2j + 1$ for some $k, j \in \mathbb{Z}$. Then

$$3(2j + 1)^2 = (2k + 1)^2.$$

Expanding and simplifying,

$$\begin{aligned} 12j^2 + 12j + 3 &= 4k^2 + 4k + 1, \\ 6j^2 + 6j + 1 &= 2k^2 + 2k. \end{aligned}$$

The left side of the last equality is odd, while the right side is even, a contradiction.

Case 2: p and q have opposite parity. Then p^2 and q^2 have opposite parity. Since $3q^2$ has the same parity as q^2 , it cannot equal p^2 , a contradiction.

Case 3: p and q are both even. Then p and q have a common factor of 2, contradicting the assumption that they are relatively prime.

These cases exhaust all possible parities of p and q . Therefore, $\sqrt{3}$ is irrational. ■

Theorem 4. $\sqrt[3]{5 - \sqrt{3}}$ is irrational.

Proof. Suppose, for the sake of contradiction, that $\sqrt[3]{5 - \sqrt{3}}$ is rational. Then

$$\sqrt[3]{5 - \sqrt{3}} = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Cubing both sides gives

$$5 - \sqrt{3} = \frac{p^3}{q^3}.$$

Rearranging,

$$\sqrt{3} = \frac{5q^3 - p^3}{q^3}.$$

Since $5q^3 - p^3 \in \mathbb{Z}$ and $q^3 \in \mathbb{N}$, this would make $\sqrt{3}$ rational, contradicting Lemma 3. Therefore, $\sqrt[3]{5 - \sqrt{3}}$ is irrational. ■

5. Problem 3.6(b)

Theorem 5. For every $n \in \mathbb{N}$ and all $a_1, a_2, \dots, a_n \in \mathbb{R}$,

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|.$$

Proof. Let $P(n)$ be the statement that, for every $a_1, a_2, \dots, a_n \in \mathbb{R}$,

$$|a_1 + a_2 + \dots + a_n| \leq |a_1| + |a_2| + \dots + |a_n|.$$

We proceed by induction on n .

Base case. For $n = 1$ and any $a_1 \in \mathbb{R}$,

$$|a_1| \leq |a_1|.$$

Thus $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Let $a_1, a_2, \dots, a_{m+1} \in \mathbb{R}$. By the induction hypothesis,

$$|a_1 + a_2 + \dots + a_m| \leq |a_1| + |a_2| + \dots + |a_m|. \quad (3)$$

Adding $|a_{m+1}|$ to both sides of (3) gives

$$|a_1 + a_2 + \dots + a_m| + |a_{m+1}| \leq |a_1| + |a_2| + \dots + |a_m| + |a_{m+1}|.$$

By the triangle inequality,

$$|(a_1 + a_2 + \dots + a_m) + a_{m+1}| \leq |a_1 + a_2 + \dots + a_m| + |a_{m+1}|.$$

Therefore, by transitivity,

$$|a_1 + a_2 + \dots + a_m + a_{m+1}| \leq |a_1| + |a_2| + \dots + |a_m| + |a_{m+1}|.$$

Thus $P(m+1)$ holds, and the result follows by induction. ■

6. Bounds table for Question 4.1

The upper- and lower-bound entries below give all real bounds for each set.

Set	Upper bounds	Lower bounds	Maximum	Minimum	Supremum	Infimum
(h)	DNE	$\{x \in \mathbb{R} : x \leq 2\}$	DNE	2	DNE	2
(i)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq 0\}$	1	0	1	0
(j)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq \frac{2}{3}\}$	DNE	$\frac{2}{3}$	1	$\frac{2}{3}$
(k)	DNE	$\{x \in \mathbb{R} : x \leq 0\}$	DNE	0	DNE	0
(n)	$\{x \in \mathbb{R} : x \geq \sqrt{2}\}$	$\{x \in \mathbb{R} : x \leq -\sqrt{2}\}$	DNE	DNE	$\sqrt{2}$	$-\sqrt{2}$
(r)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq 1\}$	1	1	1	1
(w)	$\{x \in \mathbb{R} : x \geq \frac{\sqrt{3}}{2}\}$	$\{x \in \mathbb{R} : x \leq -\frac{\sqrt{3}}{2}\}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$

7. Problem 4.7

Theorem 6 (Part (a)). *Let S and T be nonempty bounded subsets of $\mathbb{R}$. If $S \subseteq T$, then*

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

Proof. Suppose $S \subseteq T$.

Supremum case. Suppose, for the sake of contradiction, that $\sup S > \sup T$. Since $\sup T$ is an upper bound for T , every $s \in S$ also satisfies $s \leq \sup T$. Thus $\sup T$ is an upper bound for S . But $\sup T < \sup S$, contradicting the fact that $\sup S$ is the least upper bound for S . Hence $\sup S \leq \sup T$.

Infimum case. Suppose, for the sake of contradiction, that $\inf S < \inf T$. Since $\inf T$ is a lower bound for T , every $s \in S$ also satisfies $\inf T \leq s$. Thus $\inf T$ is a lower bound for S . But $\inf T > \inf S$, contradicting the fact that $\inf S$ is the greatest lower bound for S . Hence $\inf T \leq \inf S$.

Since S is nonempty, choose $s \in S$. The definitions of infimum and supremum give $\inf S \leq s \leq \sup S$, so $\inf S \leq \sup S$. Combining the three inequalities yields

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

■

Theorem 7 (Part (b)). *Let S and T be nonempty bounded subsets of $\mathbb{R}$. Then*

$$\sup(S \cup T) = \max\{\sup S, \sup T\}.$$

Proof. Let $x = \max\{\sup S, \sup T\}$ and $\alpha = \sup(S \cup T)$. Since $S \subseteq S \cup T$ and $T \subseteq S \cup T$, part (a) gives

$$\sup S \leq \alpha \quad \text{and} \quad \sup T \leq \alpha.$$

Therefore $x \leq \alpha$.

Suppose, for the sake of contradiction, that $x < \alpha$. Then $0 < \alpha - x$, so

$$0 < \frac{\alpha - x}{2} < \alpha - x.$$

Adding x gives

$$x < x + \frac{\alpha - x}{2} < \alpha.$$

Since $x \geq \sup S$ and $x \geq \sup T$, x is an upper bound for both S and T . Consequently, $x + \frac{\alpha - x}{2}$ is an upper bound for $S \cup T$. This contradicts the fact that α is the least upper bound for $S \cup T$. Thus $x = \alpha$, or

$$\sup(S \cup T) = \max\{\sup S, \sup T\}.$$

■

8. Problem 4.8(b)

Theorem 8. *Let S and T be nonempty subsets of $\mathbb{R}$ such that*

$$s \leq t \quad \text{for all } s \in S \text{ and } t \in T.$$

Then $\sup S \leq \inf T$.

Proof. Let S and T be nonempty subsets of $\mathbb{R}$ with the following property:

$$s \leq t \quad \text{for all } s \in S \text{ and } t \in T.$$

Suppose for the sake of contradiction that $\sup(S) > \inf(T)$. Thus, $\inf(T)$ is not an upper bound for S . Hence, there exists an $s \in S$ such that $\inf(T) < s < \sup(S)$. Let s_1 be such s . Now we have two cases:

Case 1. s_1 is a lower bound for T , in which case, s_1 is a greater lower bound for T than $\inf(T)$, which is a contradiction.

Case 2. $s_1 \in T$. Since $s_1 \leq t$ for all $s \in S$ and $t \in T$, then $\min(T) = s_1$. Since s_1 is the minimum of T , then $s_1 = \inf(T)$, contradicting $\inf(T) < s_1$.

Case 3. s_1 is an upper bound for T , in which case, for all $t \in T$, $t \leq s_1$. If $t < s_1$ for any $t \in T$, then we contradict $s \leq t$ for all $s \in S$ and $t \in T$. Thus, $t = s_1$ for all $t \in T$. However, since $s_1 < \sup(S)$, then there exists $s_2 \in S$ such that $s_1 < s_2 < \sup(S)$. Thus, $s_2 \in S$ such that $s_2 > t$ for all $t \in T$, contradicting our supposition that for all $s \in S$ and for all $t \in T$, $s \leq t$.

Hence, we've exhausted all cases with contradictions.

Therefore, $\sup(S) \leq \inf(T)$. ■